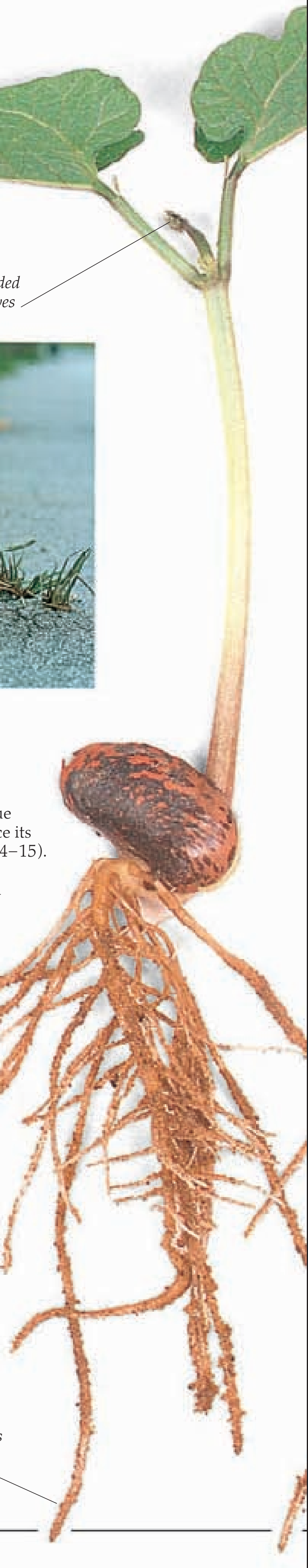
A plant is born

A SEED IS A TINY LIFE-SUPPORT PACKAGE. It contains a plant embryo—the basic parts from which the seedling will develop—together with a supply of food. The food is needed to keep the embryo alive and fuel the process of germination (growth). It is either packed around the embryo, in an endosperm, or stored in special seed leaves, known as cotyledons. For weeks, months, or even years, the seed may remain inactive. But then, when the conditions are right, it suddenly comes alive and begins to grow. During germination the seed absorbs water, the cells of the embryo start to divide, and eventually the seed case, or testa, breaks open. First, the beginning of the root system, or radicle, sprouts and grows downward, followed rapidly by the shoot, or plumule, which will produce the stem and leaves.

TINY BUT STRONG
When plants grow, they can exert great pressure. Some seedlings can easily push through the surface of a new road.

1 GETTING GOING
The seed of a scarlet runner will germinate only if it is dark and damp. First, the skin of the seed splits. The beginning of the root system, the radicle, appears and starts to grow downward. Shortly after this, a shoot appears, initially bent double with its tip buried in the seed leaves. This shoot, or plumule, will produce the stem and leaves.

2 REACHING FOR THE LIGHT
As the plumule grows longer, it breaks above ground. Once it is above the soil, it straightens up toward the light, and the first true leaves appear. In the scarlet runner, the seed leaves stay buried. This is called hypogeal germination. In plants such as the sunflower, the seed leaves are lifted above ground, where they turn green and start to produce food for the seedling. This is known as epigeal germination.



3 HARNESSING THE SUN
With the opening of the first true leaves, the seedling starts to produce its own food by photosynthesis (pp. 14–15). Until this time, its growth is fueled entirely by the food reserves stored in the seed leaves.